

Sharmista Shastry

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EDUCATION

Columbia University

M.S. in Computer Science (Machine Learning Track) GPA: 3.9/4.0

New York, NY

Dec 2024

Vellore Institute of Technology

B.Tech. in Electronics and Communication Engineering GPA: 4.0/4.0

Vellore, India

June 2021

PROFESSIONAL EXPERIENCE

Coinbase

Software Engineer

New York, NY

Dec 2024

- Led the datastore migration for Python-based Airflow DAGs and Nomad services from PostgreSQL to ClickHouse, planning dual-run verification and a clean cutover that **improved data freshness to 99%** and **unlocked faster analytics with lower operational overhead**.
- Refactored core Python services to remove redundant queries and centralize read/write paths, **improving on-time data delivery and stability** even during 2x market spikes.

Commonwealth Care Alliance

AI/ML Engineer, Intern

Boston, MA

May - Aug 2024

- Developed XGBoost-based predictive models with MLOps pipelines, enabling proactive cost management with **\$1.2M projected savings**. Built dashboards for real-time model monitoring, **reducing stakeholder time-to-insight by 40%**.

Nutanix

Software Engineer

Bangalore, India

Aug 2021 – July 2023

- Designed and built scalable microservices using Python and Go, **improving data processing performance by 25%** and **decreasing system downtime by 15%**.
- Designed and implemented automation tools for cluster lifecycle management, reducing manual intervention and **cutting average provisioning time by 30%**.

Software Engineer, Intern

Jan 2021 – June 2021

- Assisted in developing **Python scripts** for automating repetitive tasks related to data processing and migration, reducing manual effort by 40%.

PROJECTS

“Distributed Key/Value Store Project - Sharded Storage System”

- Designed a fault-tolerant distributed key-value store, employing consensus protocols (Paxos) for reliability in large-scale environments. Optimized throughput with sharding techniques, enhancing system scalability and robustness.
- Key Technologies:** Go, Distributed Systems, RPCs, Fault Tolerance.

“Automated Metadata Generation System Using LLMs and Vector Databases”

- Implemented a metadata generation system leveraging Retrieval-Augmented Generation (RAG) methods and a FAISS vector database to automate the creation of metadata for project documentation and code files.
- Key Technologies:** Python, LLM, RAG, Vector Databases, AWS, Langchain, Texttract, Tika, HuggingFace Transformers

“Payment Processing System”

- Developed distinct access points for employees and customers in a payment processing system for a transaction company. Implemented features such as complex payment processing, fraud detection, transaction search, and client analytics.
- Key Technologies:** React, PostgreSQL, Python, Flask and JavaScript

RESEARCH EXPERIENCE

Research Assistant

New York, NY

NASA Goddard Institute for Space Studies

- Assisted Dr. Henrique Haas in applying **machine learning models** to analyze the impact of climate variables and agricultural practices on the NYC Watershed, contributing to data-driven policy recommendations.

Columbia Research Lab

New York, NY

- Assisted Dr. David Kessler in developing **machine learning models** to identify the spinal features in infants under 30 days to reduce the failure rate in LP procedures.

TECHNICAL SKILLS

- Programming Languages:** Python, C++, Go, Java, SQL, Bash
- AI & Machine Learning:** PyTorch, TensorFlow, Scikit-learn, OpenCV, MLOps
- Frameworks & Services:** FastAPI, Flask, Django, gRPC, REST APIs, Microservices
- Databases & Storage:** ClickHouse, PostgreSQL, Snowflake, Redis, MongoDB
- Data Engineering, Big Data and Cloud:** Apache Spark, Airflow, dbt, Kafka, Hadoop, AWS, GCP, Azure, Kubernetes, Docker, CI/CD